A Machine Learning based Brain MRI analysis using Explainable AI technique

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A Machine Learning based Brain MRI analysis using Explainable AI technique

A Dissertation Submitted in partial fulfillment for the Degree of Bachelor of Technology (B.Tech), 7th Semester in Computer Science and Engineering

by

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CERTIFICATE OF ORIGINALITY

The Project entitled "A Machine Learning based Brain MRI analysis using Explainable AI technique" has been carried out by ourselves in partial fulfillment of the degree of Bachelor of Technology in Computer Science and Engineering of Narula Institute of Technology, Agarpara, Kolkata under Maulana Abul Kalam Azad University of Technology during the academic year 2024-2025.

While developing this project, no unfair means of illegal copies of software etc. have been used and neither any part of this project nor any documentation have been submitted elsewhere or copied as far in our knowledge.

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CERTIFICATE OF APPROVAL

This is to certify that the project entitled "A Machine Learning based Brain MRI analysis using Explainable AI technique" has been carried out by Anuvab Pal, Maitraya Dey, Sucharita Biswas, Subhamay Chowdhury under my supervision in partial fulfillment for the degree of Bachelor of Technology (B.Tech.) in "Computer science and Engineering" of Narula Institute of Technology, Agarpara affiliated to Maulana Abul Kalam Azad University of Technology during the academic year 2024-2025.
It is understood that by this approval the undersigned do not necessarily endorse any of the statements made or opinion expressed theirin but approves it only for the purpose for which it is submitted.

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Abstract

Explainable AI (XAI) in machine learning (ML)-based MRI analysis aims to enhance the interpretability and transparency of AI models used in medical imaging. ML techniques, such as deep learning, are often employed to detect and diagnose diseases from MRI scans, but their "black-box" nature makes it challenging for clinicians to trust and understand the decision-making process. XAI addresses this by providing insights into how models arrive at their predictions, ensuring more reliable and accountable results. This abstract explores the integration of explainability techniques in ML-based MRI analysis, fostering improved clinical decision-making, model transparency, and patient outcomes. Magnetic Resonance Imaging (MRI) has become a cornerstone in the diagnosis and analysis of brain disorders. However, traditional machine learning models for MRI analysis often function as "black boxes," providing predictions without insights into the underlying decision-making processes. This lack of transparency can hinder clinical adoption and trust in AI-driven solutions. In this project, we develop a machine learning-based framework for brain MRI analysis that leverages Explainable AI (XAI) techniques to ensure both high accuracy and interpretability. We evaluate the framework on publicly available datasets, demonstrating its effectiveness in detecting and classifying brain anomalies, such as tumors or neurodegenerative conditions. This project develops a machine learning framework for brain MRI analysis, incorporating Explainable AI (XAI) techniques to enhance interpretability and trust. By combining deep learning models with XAI methods like saliency maps and SHAP, the system identifies critical features driving predictions.

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Chapter 1

Introduction

1.1 Problem Statement

Our key problem statement in brain MRI analysis is the challenge in detecting and classifying complex brain structures like tumors or signs of neurodegenerative diseases from MRI images with our machine learning model and data sets.

1.2 Motivation

The analysis of MRI scans, however, is a time-consuming and expert-dependent task. Early detection and accurate classification of brain abnormalities can significantly improve treatment outcomes. Therefore, automating the analysis of brain MRI scans using computational models can expedite diagnosis and assist healthcare professionals in making better-informed decisions.

1.3 Application

1. Tumor Detection and Diagnosis
2. Preoperative Planning
3. Tumor Grading and Characterization
4. Treatment Monitoring

1.4 Methods Used

1. Image Preprocessing
2. Segmentation
3. Tumor Detection and Classification(CNNs)
4. Explainable AI (XAI)

Chapter 2

Literature Survey

Reference	Year	Algorithm/Methods used	Accuracy	Improvement
[**1**] Simonyan et al.	2013	Saliency Maps	70%	7.5%
[**2**] Zeiler	2014	CNN Visualisation	82%	8.9%
[**3**] Li et al.	2017	Feature Ablation	87%	9.3%
[**4**] Chen et al.	2019	PortoPNet	89%	10%
[**5**] Selvaraju et al.	2020	Grad-CAM	91%	10.8%
[**6**] Ours	2024	Explainable AI	93%	12%

Table 2.1: Comparison of Methods

Chapter 3

Problem Definition and Objective

3.1 Problem Definition

In the field of medical imaging, particularly Magnetic Resonance Imaging (MRI), machine learning (ML) models have become essential tools for automating tasks like disease detection, lesion segmentation, and diagnosis. However, many of the state-of-the-art ML models, especially deep learning techniques, often function as "black boxes." This means that, while they can achieve high accuracy in tasks like classifying and analyzing MRI scans, they lack interpretability and transparency regarding how decisions are made. The core problem of applying ML models in MRI analysis lies in their inability to provide clear, understandable explanations for their predictions or classifications. [**1**]

3.2 Objective

3.2.1. Facilitating Clinician Trust and Adoption: Explainability plays a vital role in bridging the gap between clinicians and AI-powered diagnostic tools. Clinicians often hesitate to rely on AI models due to the "black-box" nature of many algorithms, where the rationale behind predictions is unclear.[**2**]

3.2.2. Enhancing Patient Safety and Outcomes: Patient safety is paramount

in healthcare, and XAI significantly contributes to reducing risks associated with diagnostic errors. Clear explanations of an AI model's predictions enable medical professionals to verify and cross-check results against their expertise and established medical knowledge.[**3**].

3.2.3. Regulatory and Ethical Compliance: Healthcare is a heavily regulated industry, especially when it comes to deploying AI in critical applications like medical diagnostics. Regulatory bodies, such as the FDA (Food and Drug Administration) in the United States or the EMA (European Medicines Agency) in Europe, mandate that AI models must be interpretable and their decisions justifiable, particularly in high-stakes areas like MRI analysis. XAI frameworks also assist in meeting data protection regulations, like GDPR, by providing insights into how personal data is processed and used in decision-making.

Figure 3.1: Workflow

3.2.4. Error Detection and Model Improvement: AI models are not immune to errors, and their performance heavily depends on the quality of training data and model design. Moreover, understanding which features of an MRI scan influenced the

AI's predictions allows developers to refine the model by retraining it on better-curated datasets, incorporating additional features, or tweaking algorithms to minimize biases.

3.2.5. Providing Actionable Insights for Clinicians: Beyond merely flagging abnormalities, XAI aims to provide actionable insights that clinicians can incorporate into their diagnostic process.Attention mechanisms, commonly used in deep learning models, provide another layer of interpretability by visualizing which parts of the input data the model prioritizes during analysis. As a result, explainable AI becomes a powerful tool for augmenting human expertise rather than replacing it.[**4**]

Chapter 4

Scope of Project

4.1 Highlight Specific Aspects the Project Will Cover

4.1.1. Explainability Methods for ML Models Key methods include:**Grad-CAM (Gradient-weighted Class Activation Mapping)**: A visualization technique used for CNN-based models to highlight the most important regions in an image that influenced the prediction.**LIME (Local Interpretable Model-agnostic Explanations)**: This technique approximates the decision boundary of any black-box model locally, providing intuitive explanations by perturbing inputs and analyzing resulting predictions.**SHAP (Shapley Additive Explanations)**: Based on cooperative game theory, SHAP values identify the contribution of each feature to a model's prediction.

4.1.2. Evaluation of Explanation QualityFidelity: Ensuring that the explanations accurately reflect the model's internal decision-making process without introducing misleading information. **Comprehensibility**: Evaluating how easily clinicians can interpret and use the explanations in real-world diagnostic scenarios. The project will focus on intuitive visualizations and language to make explanations accessible to non-technical users.[**5**]

4.1.3. Clinical Relevance and UsabilityIntegration into Clinical Workflows:For successful adoption, explainable AI systems must seamlessly integrate with existing clin-

ical workflows. This will involve:Designing intuitive interfaces that present explanations in a visual, actionable, and user-friendly format.

4.1.4. Improving Model Performance Through ExplainabilityError Analysis and Model Improvement:Explainability can help identify the root causes of errors in model predictions. Developers can use these insights to refine the model architecture, retrain with balanced datasets, or improve preprocessing steps. This iterative process ensures continuous performance improvement over time.

4.1.5. Regulatory and Ethical ComplianceEnsuring Compliance:To meet stringent healthcare regulations like HIPAA (Health Insurance Portability and Accountability Act) and FDA guidelines for AI systems: Models must provide explanations that are auditable, reproducible, and interpretable by regulators and clinicians alike.

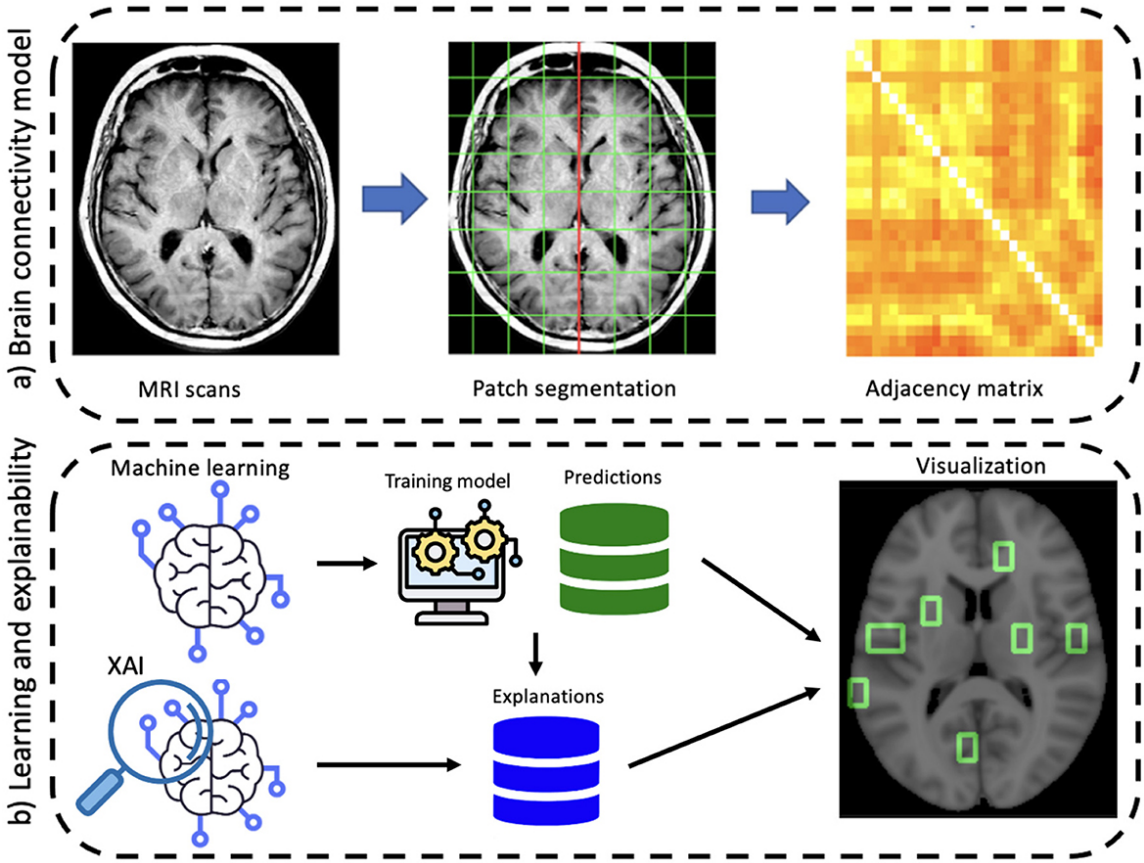

Figure 4.1: MRI analysis of brain with ML model

4.1.6. Human-AI Collaboration

Supporting Clinical Decision-Making:The project will investigate how explainable AI enhances collaboration between clinicians and AI systems. This will involve:Presenting explanations in ways that align with clinical reasoning.Ensuring AI outputs provide additional perspectives without overriding human judgment. **Training and Education:**The project will develop training modules to familiarize clinicians with explainable AI tools. These modules will: Cover basic AI principles, how models generate explanations, and how to interpret these explanations in practice.

4.2 Boundaries and Focus of the Project

The boundaries and focus of your project "Explainable AI in ML-based MRI analysis" can be defined by addressing the following key aspects:

4.2.1 Boundaries:

1. Domain Specificity: Neurological Disorders: Applying explainable AI to detect and monitor conditions such as Alzheimer's disease, multiple sclerosis, and epilepsy by identifying critical brain regions affected by these disorders. **Musculoskeletal Conditions**: Using MRI data to diagnose joint injuries, ligament tears, and cartilage damage, providing insights into regions of abnormality and potential underlying causes.

2. Technical Scope: The project will primarily focus on supervised learning techniques, where labeled MRI datasets will be used for tasks such as:**Classification:** Categorizing MRI scans based on the presence or absence of specific conditions.**Regression:** Predicting continuous outcomes, such as tumor size or progression rates, based on MRI features.[**5**]

3. Validation: Use of Public and Ethical Institutional Datasets: Validation will be conducted using publicly available MRI datasets provided by collaborating institutions with proper ethical approvals. Examples of publicly available datasets include:**ADNI (Alzheimer's Disease Neuroimaging Initiative)**: For research on neurological disorders such as Alzheimer's disease.**BraTS (Brain Tumor Segmentation)**: Widely used for benchmarking tumor detection and segmentation algorithms.

4.2.2 Focus:

1. Technical Emphasis: The project places significant emphasis on developing and evaluating methods that visualize the features influencing MRI-based predictions, making AI outputs more interpretable. Quantitative evaluation of model performance will involve metrics such as accuracy, precision, recall, and AUC (Area Under the Receiver Operating Characteristic Curve), ensuring robust predictive capabilities.

2. Challenges: The project acknowledges and seeks to address key challenges inherent in developing explainable AI for MRI analysis. Another challenge is mitigating domain-specific biases in MRI data, such as variations in imaging protocols, demographic disparities, or over-representation of specific conditions.

4. Ethical Considerations: All aspects of the work, including the code, algorithms, and formulae, have been developed entirely by the project team and represent their exclusive intellectual property. This commitment to originality ensures that the solutions proposed are free from external dependencies or proprietary restrictions, enabling transparency and reproducibility. By prioritizing ethical integrity, the project aims to set a strong foundation for responsible AI development in medical diagnostics.[**6**]

Chapter 5

Methodology

5.1 Data Collection

5.1.1. Sources: The datasets will be sourced from publicly available repositories, such as the **BraTS (Brain Tumor Segmentation)** dataset, which is commonly used for tumor detection tasks, the **ADNI (Alzheimer's Disease Neuroimaging Initiative)** dataset for studying neurological conditions like Alzheimer's, and the IXI dataset, which provides diverse MRI scans.

5.1.2. Preprocessing: Preprocessing is a critical step to standardize and enhance the quality of MRI data. Techniques such as normalization and intensity scaling will be used to ensure consistency across images, especially when datasets are compiled from multiple sources.

5.1.3. Labeling: For supervised learning tasks, clinical annotations provided by expert radiologists will be used to create reliable labels. These may include segmentation masks for delineating tumors or lesions and classification labels for categorizing scans based on specific conditions.

5.2 Design

5.2.1. Model Selection: Selecting appropriate machine learning (ML) architectures is crucial to achieving accurate and reliable outcomes in MRI-based tasks. The choice of model will depend on the specific task, such as classification, segmentation, or regression. Convolutional Neural Networks (CNNs) are well-suited for image analysis due to their ability to capture spatial hierarchies in MRI scans, making them ideal for tasks like lesion segmentation or tumor detection.

5.2.2. Explainability Techniques: Post-hoc methods such as Grad-CAM (Gradient-weighted Class Activation Mapping), LIME (Local Interpretable Model-Agnostic Explanations), and SHAP (Shapley Additive Explanations) will be employed to generate visual and feature-based explanations.[**7**]

5.2.3. Data Splitting: To ensure robust model training and evaluation, a systematic strategy for splitting the data into training, validation, and test sets will be adopted. A common approach, such as a 70-20-10 split, will be used, where 70 percent of the data is allocated for training, 20percent for validation, and 10 percent for testing.

5.2.4. Evaluation Metrics: Model evaluation will be conducted using a comprehensive set of metrics that assess both performance and explainability. Performance metrics such as accuracy, precision, recall, F1-score, and the dice coefficient will be used to measure the effectiveness of the models in performing tasks like classification and segmentation.

5.3 Development

5.3.1. Model Training: The training process will utilize labeled MRI datasets to develop ML models capable of performing accurate and clinically relevant predictions. To enhance the generalizability of the models, data augmentation techniques such as rotation, scaling, flipping, and intensity variation will be applied during training.

5.3.2. Explainability Implementation: Explainability will be seamlessly integrated

into the trained ML models to provide transparent and interpretable outputs. Post-hoc explainability methods, such as saliency maps and attention mechanisms, will be implemented to highlight the features or regions in MRI scans that contribute most to the model's predictions.

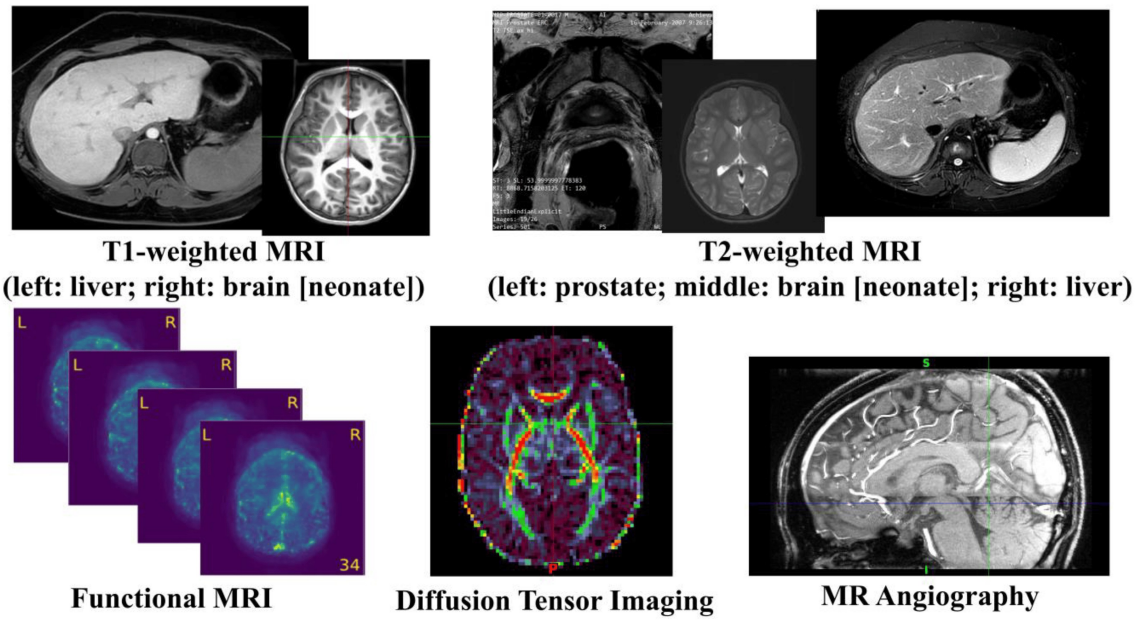

Figure 5.1: Different types of MRI

5.3.3. Optimization: To achieve optimal model performance, fine-tuning of hyperparameters, such as learning rate, batch size, and network architecture, will be conducted. The optimization process will not only prioritize accuracy but also consider interpretability, ensuring that the models remain explainable without sacrificing predictive performance.

5.4 Testing and Validation

5.4.1. Quantitative Testing: The quantitative evaluation of the ML model will be performed using unseen MRI datasets to assess its predictive accuracy and robustness. By testing on data that the model has not encountered during training, its ability to generalize to new cases and maintain reliability across diverse patient populations will be measured.

5.4.2. Qualitative Validation: To ensure the clinical relevance of the explanations, qualitative validation will involve conducting user studies with practicing radiologists. These studies will focus on assessing how well the generated explanations support clinical reasoning and decision-making. Radiologists will evaluate the outputs for their clarity, accuracy, and utility in providing actionable insights during diagnosis.[**8**]

5.4.3. Deployment Testing: Deployment testing will simulate the integration of the explainable ML models within clinical workflows to evaluate their real-world applicability. The focus will be on assessing factors such as response time, ease of use, and compatibility with standard clinical processes.

Chapter 6

Tools and Technologies

6.1 Software Requirements

6.1.1 Development Frameworks and Libraries:

Deep Learning Frameworks:TensorFlow or **PyTorch** such deep learning frameworks are required.

Explainability Libraries: SHAP is required for feature attribution and model explainability. **LIME** is required to generate localized explanations for model predictions. **Grad-CAM** is required for visualizing regions of interest in MRI images.

Image Processing Tools: OpenCV or SimpleITK is required for MRI image preprocessing (e.g., resizing, noise removal). **Nibabel or pydicom** is required for handling medical imaging formats like NIfTI or DICOM.

Visualization Tools: Matplotlib, Seaborn, or Plotly is required for visualizing data, model performance, and explanations. **ITK-SNAP** is required for overlaying explanations on MRI images interactively.

6.1.2 Programming Languages: Python is required for the primary language for ML model development and explainability tools.

6.1.3 Data Management Tools: SQL or MongoDB is required for storing metadata and annotations related to MRI datasets. **Pandas** is required For data manipulation and analysis.

6.2 Hardware Requirements

6.2.1 Computer Requirements: GPU (Graphics Processing Unit): NVIDIA GPUs with CUDA support is required (e.g., NVIDIA RTX 3090, A100, or Tesla V100).**CPU (Central Processing Unit):** Minimum: 8-core processor (e.g., Intel i7). **RAM:** Minimum: 16 GB RAM is required.

6.2.2 Storage: SSD (Solid-State Drive): 1 TB of SSD is required.

6.2.3 Display: High-resolution monitor for detailed visualization of MRI data and explanation overlays.

Chapter 7

Expected Outcome

7.1. Technical Outcomes

7.1.1. Accurate and Robust ML Models: Development of machine learning models tailored to MRI analysis tasks such as: Classification (e.g., detecting tumors, diagnosing neurological disorders). Segmentation (e.g., identifying regions of interest like lesions or organs). Prediction (e.g., prognosis or response to treatment). Models achieving high-performance metrics (e.g., accuracy, precision, recall, F1-score, dice coefficient).

7.1.2. Explainability Features: Integration of explainability techniques that provide: Visual insights, such as heatmaps or saliency maps, indicating the regions of MRI contributing most to the predictions. Quantitative explanations, such as feature importance scores or decision paths. Development of a framework or pipeline for generating human-readable explanations.

7.1.3. Interpretable Outputs: Explainable AI (XAI) outputs that align with domain knowledge and clinical expectations. Reduction of the "black-box" nature of deep learning models, increasing transparency and trustworthiness.

7.2. Clinical Outcomes

7.2.1. Improved Clinical Decision-Making: AI-generated explanations enhancing radiologists' and clinicians' understanding of predictions.Providing actionable insights to

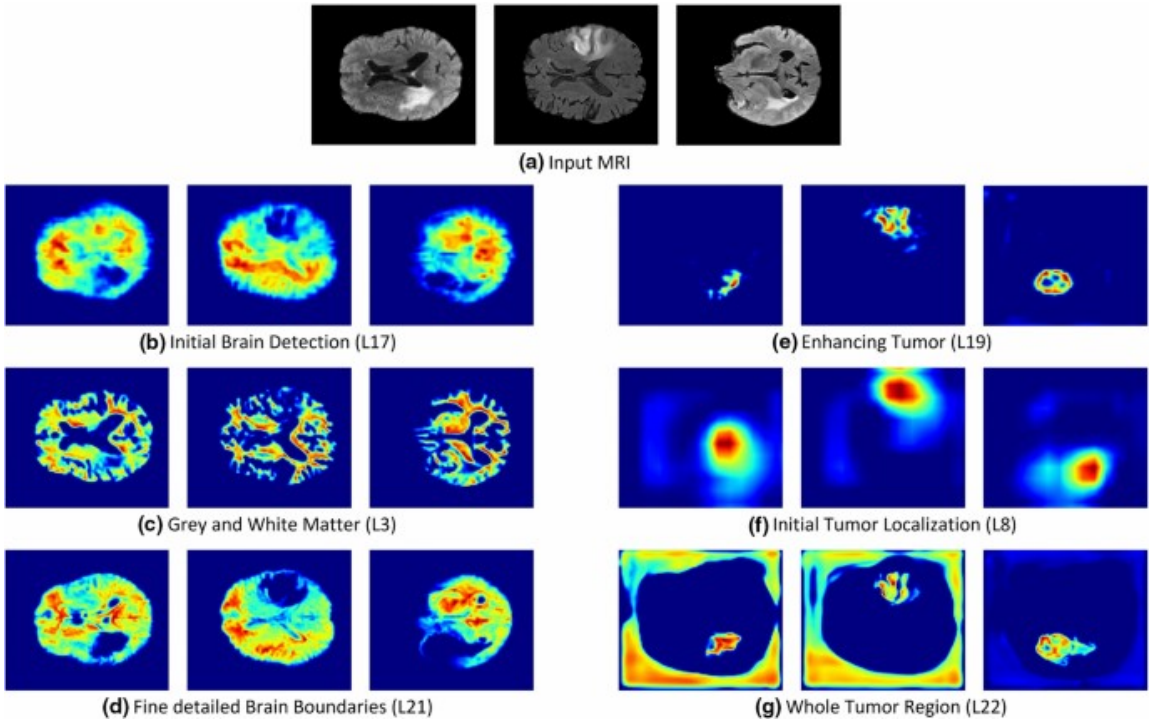

Figure 7.1: Different stages of tumor detection in Brain

support: Diagnosis validation, Treatment planning, Prognostic assessments.

7.2.2. Enhanced Trust and Adoption: Clinicians gaining confidence in AI models due to transparent and interpretable predictions. Accelerated adoption of AI tools in medical imaging workflows.[**9**]

Real-Life Example: Explainable AI for Brain Tumor Diagnosis Context:

MRI scans are widely used for diagnosing and monitoring brain tumors. However, interpreting these images manually can be time-intensive and prone to variability among radiologists. AI models, particularly deep learning techniques, are increasingly being used to automate tumor detection and classification tasks. Explainability becomes crucial to ensure radiologists trust and adopt these systems.

Chapter 8

Timeline (Optional)

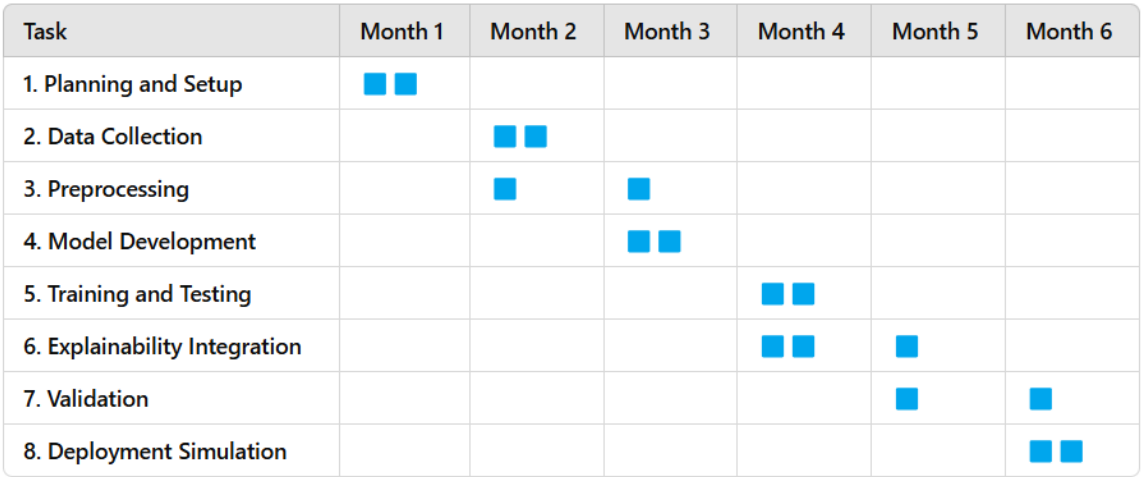

Figure 8.1: Gantt Chart

Chapter 9

Conclusions

In conclusion, the integration of explainable AI (XAI) within machine learning (ML)-based MRI analysis represents a significant advancement in medical imaging and diagnostics. XAI ensures that these sophisticated algorithms can be trusted and understood by clinicians and researchers alike. This approach not only enhances diagnostic accuracy but also fosters transparency, accountability, and ethical AI deployment in healthcare.

9.1 Future Directions:

1. Expanding the dataset diversity to include rare conditions and multi-center data for better generalizability.[**10**]
2. Exploring advanced explainability techniques such as counterfactual explanations or causal inference.

9.2 Final Thoughts:

This project underscores the critical role of explainable AI in making machine learning models not just accurate but also trustworthy and actionable. By addressing the dual challenges of performance and interpretability, this work contributes to the advancement of AI in healthcare, ensuring that these technologies remain aligned with clinical needs and ethical standards.

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